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Toward Industry 5.0: Evaluating Multimodal Virtual Human Interaction for Smart Healthcare in Simulated VR Environments

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ABSTRACT

To support the human-centric goals of Industry 5.0, this paper proposes a modular framework for constructing low-cost, high-efficiency digital humans by combining retrieval-augmented generation (RAG), large language models (LLMs), and AIGC (AI-generated content). The framework enables embodied agents capable of reliable reasoning, contextual alignment, and expressive interaction across industrial environments. As a representative application in Industry 5.0 smart healthcare, we deploy three variants—scripted, LLM-only, and LLM + RAG—in a VR-based hospital triage simulation, integrating automatic speech recognition, semantic retrieval, neural speech synthesis, facial animation, and gesture generation. A within-subject user study ($n = 45$) evaluates task accuracy, perceived naturalness, and response latency. Results show that the LLM + RAG agent significantly outperforms others in both task success (95.1%) and naturalness rating (4.61/5), as assessed via expert consensus and standardized Likert-based user ratings. These findings demonstrate that retrieval-enhanced digital humans can combine factual precision, real-time responsiveness, and multimodal expressiveness—key requirements in high-stakes, affect-sensitive domains. While healthcare is the tested use case, the architecture and evaluation protocol offer a reusable foundation for Industry 5.0 applications more broadly, including frontline services, education, and multilingual teleconsultation. The study contributes both a validated design pathway and a repeatable evaluation method for deploying scalable, trustworthy virtual agents in real-world industrial systems.

1 | Introduction

Industry 5.0 emphasizes human-centric and value-driven innovation, especially in domains like smart healthcare [1–3]. Virtual human agents are gaining traction as interfaces that mediate between patients and digital clinical systems. However, most current systems remain rule-based, limiting scalability and naturalness [4].

Large language models (LLMs) offer context-aware dialogue capabilities, yet they risk hallucination without domain grounding [5]. Retrieval-augmented generation (RAG) mitigates this by incorporating external knowledge retrieval but introduces practical challenges such as response latency, multimodal synchronization, and evaluation complexity [6, 7].

This study develops and evaluates three virtual agent variants for hospital triage: a scripted system, an LLM-only system, and an LLM + RAG system. All are implemented in a modular, real-time pipeline integrating speech recognition, semantic retrieval, language generation, TTS, facial animation, and gesture synthesis

[8, 9]. A within-subject user study ($n = 45$) assesses their performance across accuracy, naturalness, and latency in a simulated VR environment [10].

1.1 | Our Contributions Include

(i) A deployable multimodal framework for embodied healthcare agents; (ii) empirical comparison of scripted, generative, and retrieval-augmented dialogue systems; (iii) analysis of interaction trade-offs in terms of performance and responsiveness; and (iv) insights into their applicability within Industry 5.0 digital health infrastructures.

2 | Related Work

2.1 | Rule-Based vs. Data-Driven Agents

Rule-based agents rely on static scripts or finite-state machines, often limiting their flexibility and contextual awareness in dynamic environments [1]. In contrast, data-driven agents leveraging LLMs support generative and multimodal interactions with reduced manual engineering [2]. Yet, LLMs risk hallucination and inconsistency when ungrounded [3]. RAG addresses this by integrating external knowledge sources, improving factual reliability [4], although with added latency concerns [5]. However, prior work rarely offers a systematic comparison among scripted agents, LLM-only models, and RAG-enhanced systems in medical VR contexts, which this study directly addresses.

2.2 | Digital Human Modeling

Advances in neural rendering and speech synthesis now enable realistic digital humans. MuseTalk generates lip-synced animations at 30+ FPS, and DiffStyleGesture maps text to appropriate gestures [6]. Effective agents must combine human-like visuals with consistent personality and emotion [7–9]. Facial realism and natural voices enhance immersion [10, 11], yet many remain limited in emotional depth [12]. Integrating expressive cues is essential, while latency over 4 s still disrupts experience despite mitigation strategies like conversational fillers [13, 14]. Our work builds on these foundations by evaluating the entire speech–gesture pipeline in real-time clinical scenarios, rather than isolated generation tasks.

2.3 | Evaluation and User Trust

While LLM-driven digital humans have made technical strides, challenges remain in systematic evaluation and user trust. A recent survey highlights the lack of standardized frameworks for assessing multimodal interactions, making it difficult to compare systems or link design decisions to outcomes [15]. Current evaluations often emphasize subjective impressions, overlooking gesture diversity, generation speed, and ecological validity [16]. Trust in AI agents depends not just on dialogue quality, but also on transparency, reliability, and ethical alignment. Prior studies show that consistent personality and expressive coherence enhance engagement, while mismatched behavior can undermine trust [17]. Unlike previous evaluations, we use both subjective ratings and task-grounded accuracy to assess interaction reliability.

3 | Methodology

3.1 | System Overview

We adopt a modular architecture (Table 1) consisting of six components. (i) ASR uses Whisper to transcribe user speech into text (temperature is set to 0.0 to ensure deterministic decoding and consistency across medical queries); (ii) retrieval employs FAISS and sentence-BERT embeddings to obtain the top-k relevant documents from a domain-specific knowledge base; (iii) LLM dialogue relies on GPT-4 or LLaMA2 to generate responses conditioned on the user query and retrieved context; (iv) text-to-speech (TTS) synthesizes natural speech via neural TTS models; (v) facial and gesture animation uses MuseTalk for lip-synchronized facial motion and DiffStyleGesture for semantically appropriate gestures; and (vi) unity-based rendering integrates audio, facial animation and gestures into a real-time VR environment. Components communicate asynchronously to minimize bottlenecks.

The architecture supports three system variants—scripted, LLM-only, and LLM + RAG—by enabling or disabling retrieval and generation modules. In the scripted version, agent responses are selected from predefined templates based on keyword triggers. In the LLM-only version, the model generates responses directly from ASR output without external retrieval. The RAG variant augments prompts with retrieved knowledge chunks, improving factuality in domain-specific queries.

TABLE 1 | Module configurations and key parameters.

Module	Model/algorithm	Key parameters
ASR	Whisper base.zh	Sample rate: 16 kHz; chunk size: 0.05 s; temperature: 0.0
Retrieval	FAISS + sentence-BERT	Embedding dim: 384; top-k: 3; similarity threshold: 0.7
LLM dialogue	GPT-4/LLaMA-2	Max tokens: 256; temp: 0.3; system prompt includes retrieved passages
TTS	EdgeTTS/SoVITS	Voices: female (Mandarin), male (English); pitch: 0.9×; speed: 1.1×
Facial animation	MuseTalk	Res: 256 × 256; frame rate: 30 fps; latency < 0.1 s
Gesture synthesis	DiffStyleGesture	Input: text; output: 3D pose; style blending coefficient: 0.5
Depth estimation	ZoeDepth	Input res: 512 × 512; output: 3-layer depth map
vr engine	Unity 2021.3 LTS	Renderer: HDRP; audio latency compensation: 0.05 s

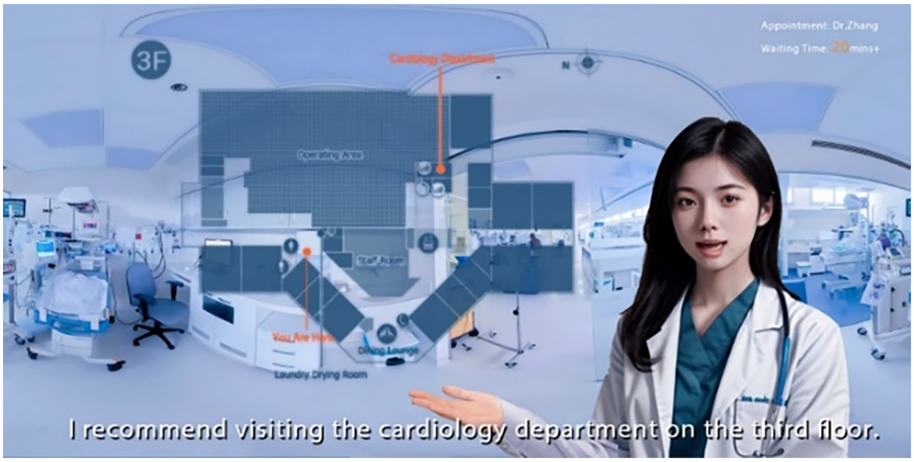

FIGURE 1 | Example of the virtual human hospital receptionist VR deploy (PC render).

During runtime, Whisper streams partial transcripts every 50 ms for early retrieval, while finalized inputs are routed to the medical triage assistant with prompt templates. Retrieval latency is reduced via embedding caching and batch FAISS lookups. To preserve multimodal coherence, the system queues speech output and synchronizes MuseTalk-driven facial animation and gesture motion.

All modules were tested on a PC with NVIDIA RTX 4090 GPU, 32 GB RAM, and a 1 Gbps LAN connection, ensuring stable performance under real-time load.

3.2 | Avatar Design

We leverage generative AI tools for creating virtual human appearances. Portraits and full-body images are synthesized using text-to-image models (e.g., RunwayML). A monocular depth estimator (ZoeDepth) processes each frame to produce depth maps for pseudo-3D rendering in VR. MuseTalk converts synthesized speech into phoneme-driven facial animations, while DiffStyleGesture generates upper-body gestures aligned with the semantic content. We selected MuseTalk for its high-fidelity lip sync and minimal delay (< 300 ms), enabling real-time audiovisual synchronization in VR Figure 1 is a render sample.

3.3 | Speech Synthesis and Animation

To support multimodal interaction, our system generates real-time speech and facial animation based on LLM responses. Neural TTS converts text into lifelike audio, while MuseTalk drives lip sync and facial expressions aligned to the speech waveform. Gesture and head pose animations are generated using predefined motion templates and synchronized with the speech timeline. All modules operate asynchronously to maintain responsiveness and allow scalability—key requirements in Industry 5.0 smart healthcare environments.

3.4 | Hospital Triage Assistant Scenarios

This scenario simulates a hospital receptionist. Participants describe symptoms in Mandarin or English, and the agent recommends departments accordingly. Scripted replies use fixed templates; LLM only generates free-form responses; LLM + RAG augments prompts with retrieved documents to aid understanding of vague or colloquial descriptions.

4 | Experiments and Results

4.1 | Participants and Procedure

Based on the system architecture described in Section 3, we implemented three virtual hospital triage agents—scripted, LLM-only, and LLM + RAG—integrated into an interactive VR setup. All modules were deployed on a standalone VR headset (Meta Quest Pro, Snapdragon XR2+, 12 GB RAM) connected to WiFi 6.

We recruited 45 participants (22 male, 23 female) aged 20–35 with mixed VR experience. This age range was selected to ensure consistent digital literacy while allowing for variation in prior immersion exposure. Eighteen participants reported no VR use, while 27 had varying levels of experience, enhancing ecological diversity.

Participants received a brief tutorial and completed a practice interaction with a non-study avatar. Each full session consisted of three blocks (scripted, LLM-only, LLM + RAG) presented in randomized order. In each block, participants described symptoms to a virtual agent, which responded in real time. For each interaction, system input, output, and response timing were recorded. Task success was judged by two medical experts based on alignment with the intended department and symptom logic; disagreements were resolved via consensus.

After each interaction, participants completed a structured survey assessing perceived naturalness and trust using a 5-point Likert scale(1 = *strongly disagree*, 5 = *strongly agree*). They could also provide free-form comments. This evaluation pipeline reflects a reusable assessment model for testing multimodal agents in applied healthcare contexts.

4.2 | Metrics and Statistical Analysis

We evaluated three key metrics: task accuracy, perceived naturalness and trustworthiness, and response latency. Task accuracy was calculated as the proportion of tasks judged as successful by two medical experts, based on whether system outputs matched the correct department and relevant symptoms. Disagreements were resolved via consensus. All ± values indicate 95% confidence intervals across the participant sample, reflecting sample-level variability.

Perceived naturalness and trustworthiness were evaluated using a 5-point Likert scale. Naturalness was decomposed into five dimensions: speech fluency, facial synchronization, gesture appropriateness, emotional expressiveness, and overall realism. Ratings were collected after each interaction block and averaged across participants.

Response latency was computed as the time (in seconds) between user input completion and the start of system-generated speech. Timing was logged automatically during each trial.

To assess measurement reliability, Cronbach's α across all survey items reached 0.85, confirming the internal consistency of the subjective evaluation instrument.

System-level comparisons were conducted using repeated-measures ANOVA to account for within-subject variability, with Bonferroni correction applied to control for Type I error across multiple pairwise comparisons. This procedure ensures statistical rigor in evaluating differences across the three system architectures.

4.3 | Results

To summarize, the LLM + RAG agent achieved the highest task accuracy at 95.1% (±0.9), followed by LLM-only at 87.4% (±1.1) and scripted at 73.8% (±1.2).

In terms of response latency, the scripted agent responded fastest at 2.01 s (±0.08), while LLM-only and LLM + RAG showed slightly longer delays at 2.78 s (±0.09) and 3.49 s (±0.07), respectively. These latency differences reflect trade-offs between speed and response quality.

Repeated-measures ANOVA confirmed that system type had a significant effect on all three metrics ($p < 0.001$), with the LLM + RAG agent consistently outperforming the other systems in both objective and subjective evaluations. No significant interaction effect was found between system type and scenario, suggesting robust trends across task conditions.

Beyond aggregate scores, a finer-grained analysis of naturalness ratings across five dimensions is shown in Table 2. The LLM + RAG system received the highest ratings across all dimensions, especially in speech fluency and gesture appropriateness, indicating the benefits of retrieval-enhanced contextual reasoning and dynamic expressiveness.

To evaluate the internal consistency of the five naturalness dimensions, Table 3 summarizes their Cronbach's α values and overall ratings aggregated across all systems and participants. All five dimensions exceeded $\alpha > 0.8$, confirming the reliability of the measurement instrument.

TABLE 2 | Naturalness ratings per system and dimension (mean ± 95% CI, across participants).

Dimension	Scripted	LLM-only	LLM + RAG
Speech fluency	3.85 ± 0.15	4.15 ± 0.13	4.50 ± 0.10
Facial synchronization	3.78 ± 0.18	4.10 ± 0.14	4.35 ± 0.11
Gesture appropriateness	3.60 ± 0.20	4.00 ± 0.15	4.25 ± 0.12
Emotional expressiveness	3.55 ± 0.18	3.95 ± 0.14	4.25 ± 0.13
Overall realism	3.80 ± 0.16	4.10 ± 0.13	4.40 ± 0.11

TABLE 3 | Aggregated ratings and internal consistency across all systems.

Survey item	Cronbach's α	Mean rating (±95% CI)
Speech fluency	0.85	4.35 ± 0.12
Facial synchronization	0.88	4.22 ± 0.11
Gesture appropriateness	0.86	4.08 ± 0.10
Emotional expressiveness	0.83	4.01 ± 0.13
Overall realism	0.87	4.15 ± 0.12

TABLE 4 | Performance metrics across dialogue engines (mean ± 95% CI).

Metric	Scripted	LLM-Only	LLM + RAG
Task accuracy (%)	73.8 ± 1.2	87.4 ± 1.1	95.1 ± 0.9
Naturalness (1–5)	3.25 ± 0.13	4.12 ± 0.11	4.61 ± 0.10
Latency (s)	2.01 ± 0.08	2.78 ± 0.09	3.49 ± 0.07

A summary of performance across all three metrics—task accuracy, naturalness, and latency—is presented in Table 4.

Qualitative feedback further supported these trends: participants appreciated the LLM + RAG agent's ability to deliver contextual explanations and natural gestures, with several noting that the receptionist "felt like a real person." Conversely, the scripted system was criticized for symptom misclassification and lack of expressiveness. Some noted TTS mispronunciations and occasional latency in the LLM-based systems, though these were generally tolerated in light of the improved interaction quality.

These findings underscore the added value of retrieval-augmented multimodal interaction in complex healthcare scenarios, and point to future opportunities for optimizing latency and prosodic alignment.

5 | Discussion

The results demonstrate that RAG significantly enhances both task accuracy and perceived naturalness compared to LLM-only and scripted approaches. This supports findings in related conversational systems research [14, 16] but extends them to a multimodal, VR-based medical setting. Unlike prior work focusing on single-turn or text-only dialogues [17], our study highlights the benefits of RAG in sustained, embodied interaction, particularly in scenarios with ambiguous or colloquial user input.

While LLM-only agents offered good naturalness and latency, their lower accuracy highlights weak domain grounding without external knowledge. Scripted agents were fast but scored lowest overall, reflecting the limits of rule-based systems. Retrieval-enhanced dialogue improved context understanding and informativeness.

However, several limitations merit attention. First, the sample consisted mainly of university-aged participants (20–35), which may limit generalizability to older users or more diverse clinical settings [3]. Second, the VR setup required relatively high computing and network resources, which could affect deployment feasibility in lower-resource environments. Third, although task accuracy and naturalness were assessed with validated metrics, other aspects such as emotional trustworthiness, long-term engagement, or dialectal variation were not systematically evaluated.

6 | Conclusion

This study evaluated three virtual human architectures—Scripted, LLM-only, and LLM + RAG—within a VR-based hospital triage simulation. The LLM + RAG agent demonstrated the highest task accuracy and naturalness ratings, confirming the effectiveness of retrieval augmentation for domain-grounded, real-time interaction.

We introduced a modular development framework that combines language models, semantic retrieval, neural speech synthesis, facial and gesture generation, enabling expressive yet cost-efficient digital humans. Unlike traditional scripted systems or opaque LLM-only pipelines, our approach balances controllability with flexibility—supporting accurate, human-aligned responses in dynamic environments. Our evaluation framework offers a standardized, reusable process for multimodal agent assessment. Task success was determined by medical experts via consensus, while naturalness and trust were rated using a five-dimensional Likert scale. Reliability was confirmed via Cronbach's α (0.85), ensuring internal consistency across perceptual ratings. This structure supports fair comparison across systems and scenarios.

Although this study focused on healthcare, the framework can generalize to broader Industry 5.0 applications, including eldercare, emergency response, education, and multilingual services. Beyond VR, LLM + RAG agents can serve as foundational interfaces for human-AI teaming in physical robots, kiosks, or IoT platforms—where contextual reasoning, latency control, and emotional awareness are critical.

Future work will explore deployment in diverse environments and populations, and develop standardized benchmarks (e.g., symptom disambiguation accuracy, affective response alignment) to enhance replicability. Overall, this study advances the design of trustworthy, multimodal agents for next-generation industrial services.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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